

JOSEPH E. GEIS

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github.com/josephegeis [in linkedin.com/in/josephegeis](https://www.linkedin.com/in/josephegeis)

EDUCATION & SKILLS

Brigham Young University

December 2028

B.S. in Mathematics: Applied & Computational Mathematics; Minor in Computer Science

GPA: 3.96 ACT: 36

Heritage Scholar (full-tuition); Dean's List, College of Computational, Mathematical, and Physical Sciences

JavaScript Node.js, TypeScript, React, TanStack, Express, Next.js, Knex.js, Vite, Zod, Socket.IO

Python FastAPI, NumPy, pandas, matplotlib, PyTorch

Systems/Infra PostgreSQL, Rust, Java, Linux, Docker, Git, GitHub Actions, Caddy, gRPC

Spoken Languages English, Mandarin Chinese (OPI: Advanced Low)

WORK EXPERIENCE

Software Engineer/Research Assistant, MATRIX Lab

March 2026 - present

Brigham Young University, Computer Science Department

Provo, UT

- Maintained a data collection React/Express application used by 90+ global contributors to gather over 2,000 hours of African speech data in over 30 languages.
- Migrated 3M+ rows of corpus data from legacy SQLite database to production-ready PostgreSQL database, improving app performance by 6x.
- Designed and developed ML TTS/translation web API and service architecture to host four in-house-developed models.
- Integrated Hugging Face SDKs to pull models from Hub and use Transformers and Tokenizers for TTS pipelines.

Teaching Assistant, Computer Science 111

September 2022 - April 2023 · January 2026 - April 2026

Brigham Young University, Computer Science Department

Provo, UT

- Redesigned the course website using frameworks like Tailwind CSS and Eleventy used by hundreds of students.
- Taught twice-weekly introductory Python & programming skills classes to 80 students.
- Analyzed gradebook data with pandas and matplotlib, enabling professors to identify and support students in need.

PROJECTS

mathematica-mcp (github.com/josephegeis/mathematica-mcp)

Python, FastMCP

- MCP server providing tools for agents to perform calculations and render graphics with Wolfram Mathematica kernels.

buzzin (github.com/josephegeis/buzzin)

React, Express, Socket.IO

- Game buzzer built in React with thin Express.js backend server and Socket.IO for real-time communication.

swimming.rs (github.com/josephegeis/swimming.rs)

Rust

- Work-in-progress Rust library for parsing swim meet result files.
- Future aspirations include building professional cross-platform swim meet management software.

chess (github.com/josephegeis/chess)

Java: Javalin, Lanterna; MySQL

- Multiplayer chess game with HTTP/WebSockets communication, curses-like user interface, and robust unit tests.

Additional projects: josephegeis.dev, [turing-machine](#), [VECTCALC](#), [\(more on GitHub\)](#)

ADDITIONAL INFORMATION

Hobbies Swimming, Hiking

Clubs BYU Association for Computing Machinery